

Using Machine Learning to Help Students with Learning Disabilities Learn

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Abstract. The concept behind this learning modal is to connect education with technology to meet the different needs of each student. The main aim of personalized learning is to help students with disabilities.

Students with a disability often need subject matter presented through different methods, therefore it is imperative that these technological advances benefit all students with different learning styles. Machine Learning opens up new ways to help students with disabilities. Children with autism which is a neurological disorder need a personalized development system for their daily activities. Technology can play a substantial part.

The system includes 4 parts: (i) To predict the learning level of the user. (ii) Generating multimodal learning materials using web mining. (iii) User preferences are associated with the result. (iv) Personalized contents for users delineated with an intelligent interface.

Keywords: Multimodal learning material · Special Needs Children · Web mining · Machine learning

1 Introduction

Neurological disorders are diseases of the brain, spine and the nerves that connect them. Children facing neurological disorders have different growth, learning, understanding rate. To make these children learn a model is developed.

The principle inspiration driving our model is how keen and usable innovations can be explained to the kids with Neurodevelopment issue and how noteworthy materials from the Internet can help Children for better understanding. The primary point of this work is to furnish a framework that helps kids with special requirements to comprehend materials educated in the classroom with the assistance of media components (pictures and video cuts) gathered from the web. The entire framework can be isolated into three stages which are: (a) Fetch keywords that speak to the story, (b) Fetch media components from the web as indicated by child's learning level and connect them with the keywords fetched, (c) lastly apply potential filters on the recovered components to

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P. Karrupusamy et al. (Eds.): ICSCN 2019, LNDECT 39, pp. 262–269, 2020.
https://doi.org/10.1007/978-3-030-34515-0_27

evade immaterial and improper components. The proposed plan is relied upon to helps kids with an extraordinary need to take in by gathering multimodal components from web identified with their content stories. AI calculation is utilized to recognize the learning dimension of a child. Distinguished dimension is utilized to create customized learning materials that upgrade the learnability of the kids. Different refining choices have been utilized to guarantee the fittingness of the material for Children.

2 Literature Survey

The clever framework accepts the child's story as information and discovers multimodal substance from the web with story key extraction, web mining, web refining, etc. After the hunting task is performed, URL (Uniform Resource Locator) refining, space separating and web content investigation based refining strategies are utilized to evacuate undesirable materials [1–3].

The research aims to design, develop and evaluate a mobile application providing a better learning environment with the help of usable user interface design and effortless input (children's stories) system for the generation of multimodal learning materials. Children can easily acclimate with modern computing devices and learn lessons from multimodal items of Internet resources (texts, audios, videos, and images) [4, 12, 13].

A kid once determined to have learning incapacity, the kid is additionally ordered into various sorts of learning handicaps like dyslexia, dysgraphia, dyscalculia, and dysphasia. The model will likewise recognize the dimension of Learning Disability and furnish the understudy with concentrate material for improvement [5, 8].

Foreseeing and estimate mentally unbalanced Children execution by applying data mining calculations in Weka tools. In reference to that paper recognized the learning ability relies upon a few general regular qualities like language aptitude, talking expertise, composing aptitudes and so on. Besides, gathered information over those properties, connected. [6, 7] Kids' interface for seeking and pursuing has concentrated basically on two subjects: (a) Children's interface for looking and perusing (b) Multimodal Learning Materials for Special Needs Children. The framework proposes a great interface that gives favorable circumstances to both Boolean seeking and easygoing perusing [7, 10].

Data is filled with a lot of important stories and hidden patterns. These stories and pattern can be understood by data visualization and later can be used to present it to others [11].

3 Challenges Identified

A. *How web mining techniques can be used for personalized search*

The arrangement includes first concentrates the catchphrases from the plain content gave utilizing word co-events strategy. The co-event circulation between regular terms and different terms in a sentence can assist us in finding the overall significance of a term. Removed watchwords are utilized to look for the multimodal components (writings, pictures, and video cuts) and other comparable stories over the web.

B. *Can Machine learning calculation distinguish the learning extent of a youngster*

The mental imbalance kids learning aptitudes are affected by different components like consideration, script, spelling, language, perusing, composing, understanding abilities and result from the diverse kids. Tests were taken for usage. It examines the information and recognizes examples and in this manner distinguishes the dimension of understudy.

C. *Can distinguished dimension be utilized to create customized learning materials which improve the learnability of the Children*

Initially, a sample data stored in the database is used for training and testing the system. The machine-learning algorithm uses this data to cluster students based on previous related data. The algorithm which gives the best results is selected and the process is carried on. Once students are classified into various clusters, we provide them with study materials based on the level of Learning Disability which is the same for all the students in the same cluster.

D. *How acute interfaces can improve the connection of kids with extraordinary necessities*

The principal object is to configuration, create and test model for Children with unique needs offering a UI with adaptability and ease of use. For designing and developing the prototype UCD (User-Centered Design) approach was adopted. This prototype contains almost all sorts of such required elements that conform to a high standard of usability.

4 Problem Definition

They find it hard to learn and adapt things by the traditional method and see things from other people's perspectives. Personalized learning will be modified learning programs per their learning and progress curve and this can help the teacher track the improvement as well.

5 Proposed System Methodology

5.1 Flowchart

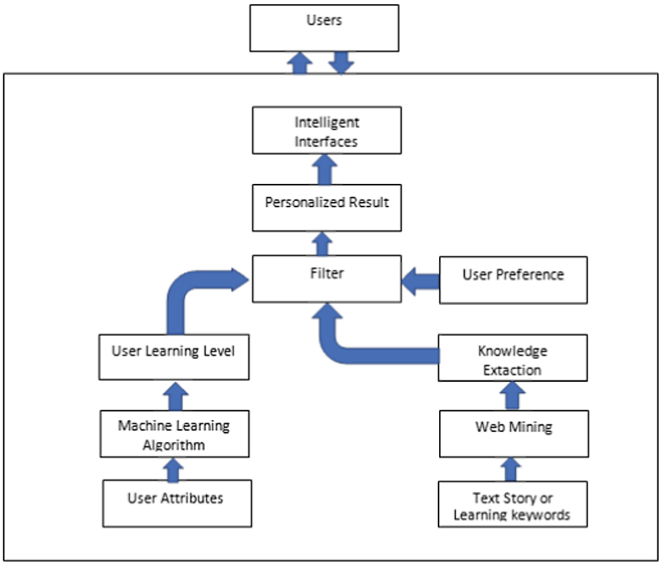


Fig. 1. Flowchart

There are three primary regions of the model as pursues:

- Web information mining, searching and separating calculation, to produce customized learning and improvement assets.
- An AI calculation to recognize the learning level and prescribe learning material for the kids.
- A multimodal, wise and usable interface for the kids thinking about their necessities (Fig. 1).

The steps in the above flowchart are explained below:

At first, the child's interfaces with the framework to scan for the required questions over the web. Here he will give a content story or learning catchphrases for tracking. Catchphrases would be separated from the narratives. The calculation utilized for watchword extraction depends on Keyword Extraction from a solitary archive utilizing Word Co-event Statistical Information calculation. Separated catchphrases that are being mined by the web then remove learning from the World Wide Web (WWW).

A separate calculation is utilized to channel the withdrawn outcomes. In our proposed framework we utilized content arrangement to distinguish whether the website page is sheltered or not for kids from where pictures and recordings will be separated. The outcome got is connected with the child's desires.

AI calculation is utilized to effectively perceive and foresee the child's level so it would help to give the child's learning materials as their dimension. First, the information is gathered from their parents and educators about learning crippled understudies. The information gathered is put away in a database in an organized organization. Information accumulation may incorporate a sort of learning inability, the dimension of handicap in different parts of talking, perusing, tuning in, language and so on. The information put away in the database is isolated for preparing and testing the framework. When the preparation is done, we start with the testing stage. When the learning dimension of an understudy is distinguished we furnish them with study materials suitable to its dimension of learning.

5.2 Proposed Algorithm

The algorithm used for keyword extraction in web data mining is Word Co-occurrence Statistical Information algorithm

Steps are as follows:

1. *Select the successive terms and dispose of high-recurrence words which are not a piece of things and action words, for example, "the", "an", "an", "in".*
2. *Clustering frequent terms with the Jensen-Shanon divergence method.*
3. *Calculate the expected probability for the term.*
4. *Calculate co-occurrence frequency for the term*
5. *Output watchwords: Choose a given number of terms having the biggest co-event recurrence esteem.*

In our proposed system the sites which the system uses to fetch images and videos are authenticated for safe use by the children using text classification.

Steps are as follows:

1. *Check URL against a rundown of genuine URLs.*

(Sites for Children' sleep time stories, learning history, book recordings for children, infant television and so on can be spared in a database table which is later used to check against the URL of the pictures and recordings recovered from the Google)

2. *Analyze the web content*

(Filtering through pictures/recordings from risky sites, and safe outcomes are brought to the child.)

In our proposed system for determining the learning level of a child, the Multilayer perceptron algorithm gives better results (Fig. 2).

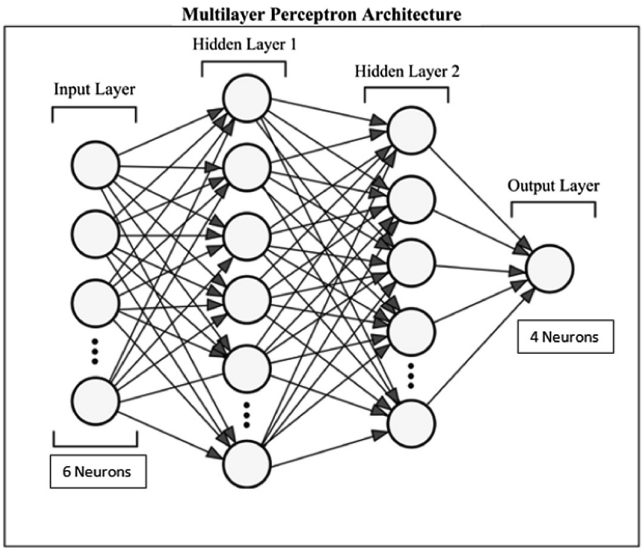


Fig. 2. Multilayer perceptron architecture

Steps followed in MLP are as follow:

1. *Determine the number of nodes in the input layer and the output layer.*
2. *Find the no. of neurons in the hidden layer by assessing the execution of the calculation.*

For most issues, we can get fair execution by setting the hidden layer setup utilizing only two guidelines:

a. Number of hidden layers equivalent to one
b. The quantity of neurons in the hidden layer is the mean of the neurons in the output and input layers

3. *Forward transmission:*

Initially, all weights in the network are randomly assigned. We now calculate the output V of the hidden layer node by applying some activation function.

$$V = f(X1 * W1 + X2 * W2 + X3 * W3 + \ldots..)$$

where f is some activation function,

X1, X2, X3….. are input values from the dataset,
W1, W2, W3……are assigned weights

Correspondingly, the yield of different neurons in the hidden layer can likewise be determined.
A hidden layer yield is taken as a contribution for the output layer. This enables us to compute the likelihood of qualities on various outputs.

Out of the first determined probabilities are a long way from the normal likelihood, so the system is considered as "false yield or false output."

4. *Now, if we want to predict the learning level of the student depending upon the input values can be done by calculating the output probability of different learning level.*

6 Performance Evaluation Parameters

The execution of the model and proficiency can be checked by an educator who assesses how well the system calculation recognizes the learning dimension of students and how well the multimodal are mapped to their learning level.

7 Conclusion

In this paper, we tend to provide every child with personalized results, as well as a training system with information of multiple users that helps to foresee the child's learning level. The proposed approach used the data sets and collected personal information of children. Personalized general materials will generate web mining techniques for efficient transmission.

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